MA1522 Linear Algebra for Computing
L2 Lesson 8



Tan Kai Meng

Semester 1, 2026/27

Question 1

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Let S, T be finite subsets of $\mathbb{R}^n$.

Determine if each of the following statements is (always) true:

a. If $S \subseteq T$, then $\text{span}(S) \subseteq \text{span}(T)$.

b. If $\text{span}(S) \subseteq \text{span}(T)$, then $S \subseteq T$.

c. $\text{span}(S \cap T) = \text{span}(S) \cap \text{span}(T)$.

d. $\text{span}(S \cup T) = \text{span}(S) \cup \text{span}(T)$.

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Subspaces

Theorem

Let V be a subset of $\mathbb{R}^n$. The following statements are equivalent:

1. $V = \text{span}(S)$ *for some (finite) subset S of $\mathbb{R}^n$.*
2. $\mathbf{0}_n \in V$ *and for all $\boldsymbol{v}_1, \boldsymbol{v}_2 \in V$ and $c \in \mathbb{R}$, $\boldsymbol{v}_1 + \boldsymbol{v}_2 \in V$ and $c\boldsymbol{v}_1 \in V$.*
3. $\mathbf{0}_n \in V$ *and for all $\boldsymbol{v}_1, \boldsymbol{v}_2 \in V$ and $c_1, c_2 \in \mathbb{R}$, $c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 \in V$.*
4. $V = \{\boldsymbol{x} \in \mathbb{R}^n \mid \boldsymbol{A}\boldsymbol{x} = \mathbf{0}_m\}$ *for some $m \in \mathbb{Z}^+$ and $m \times n$ matrix $\boldsymbol{A}$.*

When any of the above statements hold, we say that V is a **(vector) subspace** *of $\mathbb{R}^n$.*

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Determine whether each of the following sets are subspaces. If it is a subspace, express it as a linear span of vectors.

a. $V_1 = \{(p, q, p, q) \mid p, q \in \mathbb{R}\}$.

b. $V_2 = \{(a, b, c) \mid a \geq b \text{ or } b \geq c\}$.

c. $V_3 = \{(x, y, z, w) \mid 4x = 3y \text{ and } 2x = -3w\}$.

d. $V_4 = \left\{ (a, b, c, d) \;\middle|\; \det \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ a & b & c & d \end{pmatrix} = 0 \right\}$.

e. $V_5 = \{(w, x, y, z) \mid w + x = y + z\}$.

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Question 2(a,b,c): Solution

a. $V_1 = \{(p, q, p, q) \mid p, q \in \mathbb{R}\}$.

$$\begin{aligned} V_1 &= \{p(1,0,1,0) + q(0,1,0,1) \mid p, q \in \mathbb{R}\} \\ &= \operatorname{span}\{(1,0,1,0), (0,1,0,1)\}. \end{aligned}$$

b. $V_2 = \{(a, b, c) \mid a \geq b \text{ or } b \geq c\}$.

$(1,0,3), (0,2,0) \in V_2$ but $(1,0,3) + (0,2,0) = (1,2,3) \notin V_2$.
Alternatively, $(2,1,0) \in V_2$ but $-1(2,1,0) = (-2,-1,0) \notin V_2$.

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$$\begin{aligned} V_3 &= \{(x, \tfrac{4}{3}x, z, -\tfrac{2}{3}x) \mid x, z \in \mathbb{R}\} \\ &= \{x(1, \tfrac{4}{3}, 0, -\tfrac{2}{3}) + z(0,0,1,0) \mid x, z \in \mathbb{R}\} \\ &= \operatorname{span}\{(1, \tfrac{4}{3}, 0, -\tfrac{2}{3}), (0,0,1,0)\}. \end{aligned}$$

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Question 2(d,e,f): Solution

(d) $V_4 = \left\{ (a,b,c,d) \;\middle|\; \det \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ a & b & c & d \end{pmatrix} = 0 \right\}$.

$$\begin{aligned} V_4 &= \{(a,b,c,d) \mid a-c-d=0\} = \{(a,b,c,a-c) \mid a,b,c \in \mathbb{R}\} \\ &= \{a(1,0,0,1)+b(0,1,0,0)+c(0,0,1,-1) \mid a,b,c \in \mathbb{R}\} \\ &= \operatorname{span}\{(1,0,0,1),(0,1,0,0),(0,0,1,-1)\}. \end{aligned}$$

(e) $V_5 = \{(w,x,y,z) \mid w+x=y+z\}$.

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(f) $V_6 = \{(a,b,c,d) \mid ab=cd\}$.
$(1,0,2,0),(0,1,0,2) \in V_6$ but $(1,0,2,0)+(0,1,0,2)=(1,1,2,2) \notin V_6$.

Question 2(d,e,f): Solution

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$$\begin{aligned} V_4 &= \{(a,b,c,d) \mid a-c-d=0\} = \{(a,b,c,a-c) \mid a,b,c \in \mathbb{R}\} \\ &= \{a(1,0,0,1)+b(0,1,0,0)+c(0,0,1,-1) \mid a,b,c \in \mathbb{R}\} \\ &= \operatorname{span}\{(1,0,0,1),(0,1,0,0),(0,0,1,-1)\}. \end{aligned}$$

(e) $V_5 = \{(w,x,y,z) \mid w+x=y+z\}$.

$$\begin{aligned} V_5 &= \{(w,x,y,w+x-y) \mid w,x,y \in \mathbb{R}\} \\ &= \{w(1,0,0,1)+x(0,1,0,1)+y(0,0,1,-1) \mid w,x,y \in \mathbb{R}\} \\ &= \operatorname{span}\{(1,0,0,1),(0,1,0,1),(0,0,1,-1)\}. \end{aligned}$$

(f) $V_6 = \{(a,b,c,d) \mid ab=cd\}$.

$(1,0,2,0),(0,1,0,2) \in V_6$ but $(1,0,2,0)+(0,1,0,2)=(1,1,2,2) \notin V_6$.

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Question 3

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Construct a nonempty subset V of $\mathbb{R}^2$ such that V is **closed** under scalar multiplication but **not closed** under vector addition.

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If $(x, y) \in V$ and $c \in \mathbb{R}$, we have $(cx)(cy) = c^2xy = c^2(0) = 0$, so that $c(x, y) = (cx, cy) \in V$.

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$(1, 0), (0, 1) \in V$, but $(1, 0) + (0, 1) = (1, 1) \notin V$. □

Theorem

Let V be a subspace of $\mathbb{R}^n$. Let $\boldsymbol{w} \in \mathbb{R}^n$ such that $\boldsymbol{w} \notin V$. Then $\boldsymbol{w} + \boldsymbol{v} \notin V$ for all $\boldsymbol{v} \in V$.

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Proof.

Suppose for a contradiction that $\boldsymbol{w} + \boldsymbol{v} \in V$ for some $\boldsymbol{v} \in V$. Let $\boldsymbol{v}' = \boldsymbol{w} + \boldsymbol{v}$. Then $\boldsymbol{v}, \boldsymbol{v}' \in V$.

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Now

$$\boldsymbol{w} = \boldsymbol{w} + (\boldsymbol{v} - \boldsymbol{v}) = (\boldsymbol{w} + \boldsymbol{v}) - \boldsymbol{v} = \boldsymbol{v}' - \boldsymbol{v} = 1(\boldsymbol{v}') + (-1)\boldsymbol{v} \in V,$$

contradicting $\boldsymbol{w} \notin V$. □

Question 4

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Let V and W be subspaces of $\mathbb{R}^n$. Show that if $V \not\subseteq W$ and $W \not\subseteq V$, then $V \cup W$ is not a subspace (of $\mathbb{R}^n$). (Hint: use previous theorem.)

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Solution.

Let $\boldsymbol{v} \in V \setminus W$ and $\boldsymbol{w} \in W \setminus V$.

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Solution.

Let $\boldsymbol{v} \in V \setminus W$ and $\boldsymbol{w} \in W \setminus V$.
Then $\boldsymbol{v} + \boldsymbol{w} \notin V$ since $\boldsymbol{v} \in V$ and $\boldsymbol{w} \notin V$ by previous theorem.

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Then $\boldsymbol{v} + \boldsymbol{w} \notin V$ since $\boldsymbol{v} \in V$ and $\boldsymbol{w} \notin V$ by previous theorem.
Similarly, $\boldsymbol{v} + \boldsymbol{w} \notin W$ since $\boldsymbol{v} \notin W$ and $\boldsymbol{w} \in W$.

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Similarly, $\boldsymbol{v} + \boldsymbol{w} \notin W$ since $\boldsymbol{v} \notin W$ and $\boldsymbol{w} \in W$.
Thus $\boldsymbol{v} + \boldsymbol{w} \notin V \cup W$.

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Thus $\boldsymbol{v} + \boldsymbol{w} \notin V \cup W$.
But $\boldsymbol{v}, \boldsymbol{w} \in V \cup W$. Hence $V \cup W$ is not a subspace. □

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Similarly, $\boldsymbol{v} + \boldsymbol{w} \notin W$ since $\boldsymbol{v} \notin W$ and $\boldsymbol{w} \in W$.
Thus $\boldsymbol{v} + \boldsymbol{w} \notin V \cup W$.
But $\boldsymbol{v}, \boldsymbol{w} \in V \cup W$. Hence $V \cup W$ is not a subspace. □

Theorem

Let V and W be subspaces of $\mathbb{R}^n$. Then $V \cup W$ is a subspace if and only if $V \subseteq W$ or $W \subseteq V$.

Next Lesson ...

We will explore **Sections 3.4–3.5**.
Please watch the video clips for these sections before then.